

cfRNA-BrainTrace      

Bioinformatics Application Note       2026-06-19

S1 ☐ ☐ ☐ ☐ ☐ ☐

S2 ☐ ☐ ☐ ☐ ☐ ☐

S3 ☐ ☐ ☐ ☐ ☐ ☐

S4 ☐ ☐ ☐ ☐

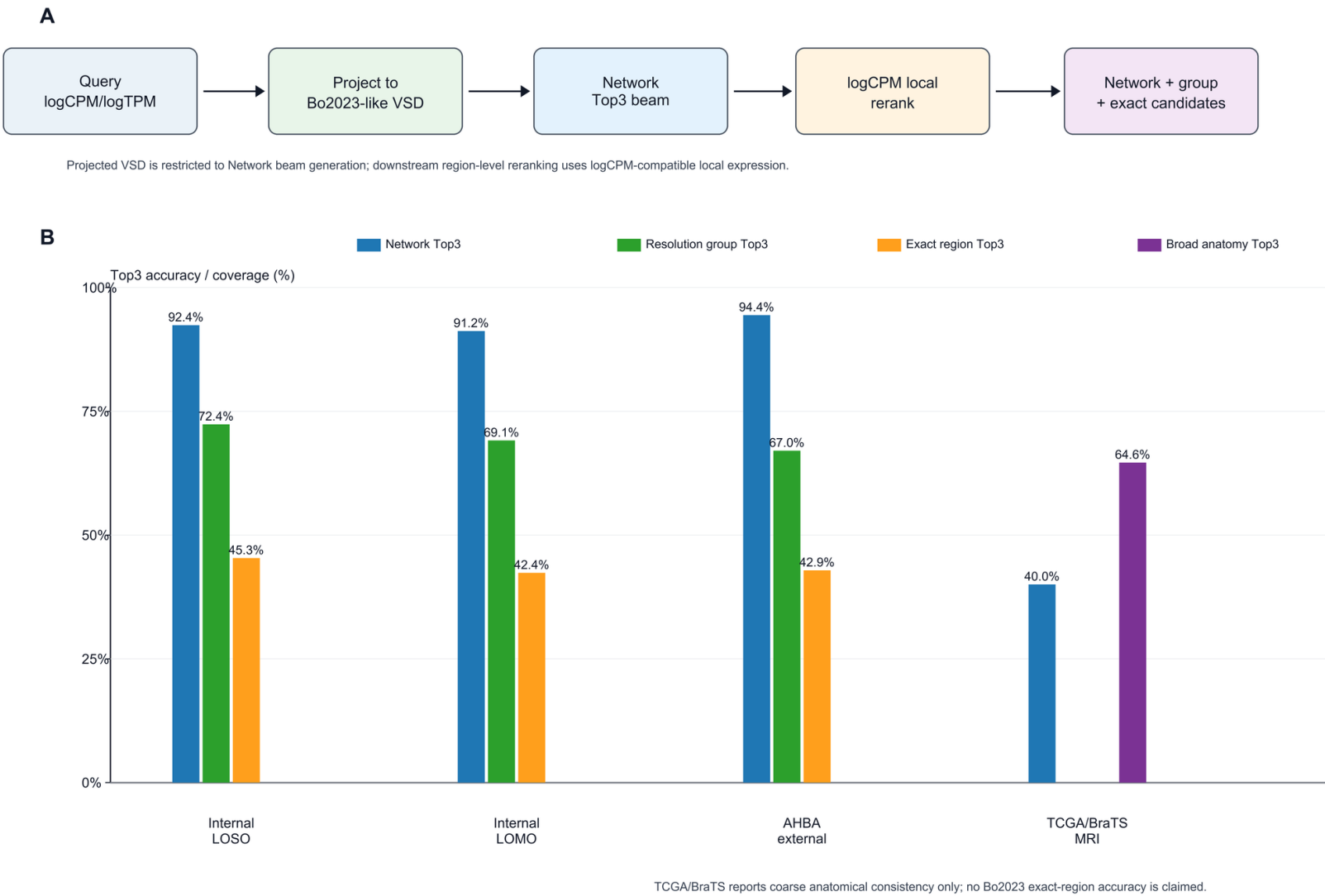
S1 ☐ ☐ ☐ ☐ ☐ ☐

S2 □□□□□□□

S3 ☐ ☐ ☐ ☐ ☐ ☐

AHBA ☐ ☐ hybrid route ☐ Network Top1/Top3 ☐ 74.68%/94.42% ☐ resolution-group Top1/Top3 ☐ 36.26%/67.03% ☐ exact-region Top1/Top3 ☐ 24.18%/42.86% ☐ Hybrid exact Top3 ☐ ☐ logCPM baseline ☐ 30.77% ☐ projected-VSD-only ☐ 29.67% ☐ TCGA/BraTS ☐ hybrid Network Top3 ☐ 40.00% ☐ broad-anatomy Top3 ☐ 64.62% ☐ ☐ ☐ coarse anatomical consistency ☐ GSE189919 ☐ ☐ 15,622/21,668 projector genes ☐ ☐ 72.10% gene-space coverage ☐ ☐ ☐ projection feasibility ☐ ☐ ☐ source-localization accuracy ☐

cfRNA-BrainTrace three-tier Application Note summary



Supplementary Table 1. TableS1_internal_validation_design

Validation	Data	Held-out unit	Route tested	Reported endpoints
Network LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank
Network LOMO	Bo2023 macaque brain RNA-seq	One monkey	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank
Formal three-tier LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected-VSD Network beam plus logCPM group/exact rerank	Network, resolution-group and exact-region Top1/Top3
Formal three-tier LOMO	Bo2023 macaque brain RNA-seq	One monkey	Fold-local Network, group and exact components	Network, resolution-group and exact-region Top1/Top3

Supplementary Table 2. TableS2_internal_validation_results

Dataset	Route	Endpoint	Top1	Top3	Median true-rank	Interpretation
Bo2023 LOSO	Projected VSD Network	Network	58.00%	91.58%	1.0	Supports projected-VSD Network beam
Bo2023 LOMO	Projected VSD Network	Network	53.72%	91.33%	1.0	Supports donor-level Network beam
Bo2023 LOSO	Formal hybrid	Network	58.35%	92.38%	1.0	Primary endpoint
Bo2023 LOSO	Formal hybrid	Resolution group	44.47%	72.36%	2.0	Main region-level endpoint
Bo2023 LOSO	Formal hybrid	Exact region	22.48%	45.33%	4.0	Exploratory ranking
Bo2023 LOMO	Formal hybrid	Network	57.75%	91.21%	1.0	Cross-monkey support
Bo2023 LOMO	Formal hybrid	Resolution group	41.38%	69.09%	2.0	Main region-level endpoint
Bo2023 LOMO	Formal hybrid	Exact region	22.17%	42.36%	5.0	Exploratory ranking

Supplementary Table 3. TableS3_external_validation_design

Dataset	Sample type	n	Truth label type	Allowed conclusion
AHBA	Human normal brain RNA-seq	242 total; 233 supported; 91 exact-evaluable	Mapped anatomical labels	Cross-species mapped-label validation
TCGA/BraTS	Glioma tissue RNA-seq with MRI-derived labels	65 patients	Human imaging labels	Coarse anatomical consistency only
GSE189919	External count matrix	51 samples	No patient-level anatomical truth	Projection feasibility only

Supplementary Table 4. TableS4_external_validation_results

Dataset	Route	Endpoint	Top1	Top3	Conclusion
AHBA	Hybrid	Network	74.68%	94.42%	Strong mapped-label Network support
AHBA	Hybrid	Resolution group	36.26%	67.03%	Moderate group-level support
AHBA	Hybrid	Exact region	24.18%	42.86%	Only exact-evaluable mapped labels
AHBA	logCPM baseline	Exact region	17.58%	30.77%	Below hybrid exact Top3
AHBA	Projected VSD only	Exact region	10.99%	29.67%	Below hybrid exact Top3
TCGA/BraTS	Hybrid	Network	15.38%	40.00%	Coarse consistency only
TCGA/BraTS	Hybrid	Broad anatomy	13.85%	64.62%	Coarse consistency only
GSE189919	Projection feasibility	Projector gene overlap	15622/21668	72.10%	No accuracy claim

Supplementary Table 5. TableS5_figure_table_index

Item	Content	Use in manuscript
Figure 1	Formal route diagram and validation summary	Main Application Note figure
Supplementary Figure S1	Main figure source artwork reproduced in the supplementary PDF	Supplementary visual reference
Table S1	Internal validation design	Documents LOSO/LOMO operations
Table S2	Internal validation results	Reports Network, group and exact-region metrics
Table S3	External validation design	Documents label support and allowed conclusions
Table S4	External validation results	Reports AHBA, TCGA/BraTS and GSE189919 outcomes

Supplementary Table 6. TableS6_claim_boundaries

Avoid	Use
Projected VSD creates a new Bo2023 atlas.	Only query profiles are projected for Network beam generation.
Projected VSD is best for exact-region inference.	Projected VSD supports Network beam; logCPM supports downstream reranking.
TCGA/BraTS validates Bo2023 exact regions.	TCGA/BraTS supports coarse anatomical consistency.
GSE189919 validates accuracy.	GSE189919 verifies projection feasibility only.
Exact Top1 is deterministic localization.	Exact outputs are exploratory local candidate rankings.